

V07 ACTIVE FURNISHINGS | O (MAX : 2 PT)

Intent : Encourage movement, postural breaks and switching and discourage prolonged sitting or standing at stationary workstations through active workstations.

Summary : This WELL feature requires projects to provide ample active workstations, such as a sit-stand or treadmill desk.

Issue : Sedentary behavior has been linked to numerous negative health outcomes, including obesity, type 2 diabetes, cardiovascular and metabolic risks and premature mortality.¹⁻⁵ Sedentary behavior also poses health risks, despite activity levels, and may even negate the positive health effects associated with physical activity.^{1,6-8}

Solutions : Active workstations have grown in popularity in commercial office environments in recent years. Active workstations are effective at decreasing time spent sitting, thereby increasing energy expenditure.⁹⁻¹³ Studies do not suggest there is an impact on productivity for sit-stand or treadmill desks with more mixed findings for bicycle desks.^{11,12,14-18} Evidence further suggests that offering active workstations along with education, prompts and/or behavior change counseling may support sustained behavior change and further reduce sitting time.¹⁹⁻²¹

PART 1 PROVIDE ACTIVE WORKSTATIONS (MAX : 2 PT)

For Office Spaces:

Active workstations meet the following requirements:

a.

Enable users to sit/stand or obtain light physical activity during use through one of the following:

- i. Manual or electric height adjustments for work surface.
- ii. Treadmill component.
- iii. Stationary bicycle component.
- iv. Step machine component.

b. Are provided in one of the following quantities:

Tier	Active Workstation Quantity	Points
1	At least 50% of workstations	1
2	At least 90% of workstations	2

REFERENCES

1. Hamilton MT, Healy GN, Dunstan DW, Zderic TW, Owen N. Too Little Exercise and Too Much Sitting: Inactivity Physiology and the Need for New Recommendations on Sedentary Behavior. *Curr Cardiovasc Risk Rep.* 2008;2(4):292-298. doi:10.1007/s12170-008-0054-8
2. Owen N, Salmon J, Koohsari MJ, Turrell G, Giles-Corti B. Sedentary behaviour and health: mapping environmental and social contexts to underpin chronic disease prevention. *Br J Sports Med.* 2014;48(3):174-177. doi:10.1136/bjsports-2013-093107
3. Young DR, Hivert M-F, Alhassan S, et al. Sedentary Behavior and Cardiovascular Morbidity and Mortality: A Science Advisory From the American Heart Association. *Circulation.* 2016;134(13):e262-79. doi:10.1161/CIR.0000000000000440
4. Chau JY, Grunseit AC, Chey T, et al. Daily Sitting Time and All-Cause Mortality: A Meta-Analysis. *Gorlova OY, ed. PLoS One.* 2013;8(11):e80000. doi:10.1371/journal.pone.0080000
5. Patterson R, McNamara E, Tainio M, et al. Sedentary behaviour and risk of all-cause, cardiovascular and cancer mortality, and incident type 2 diabetes: a systematic review and dose response meta-analysis. *Eur J Epidemiol.* 2018;1:1-19. doi:10.1007/s10654-018-0380-1
6. Owen N, Healy GN, Matthews CE, Dunstan DW. Too Much Sitting: The Population-Health Science of Sedentary Behavior. *Exerc Sport Sci Rev.* 2010;38(3):105-113. doi:10.1097/JES.0b013e3181e373a2
7. Ekelund U, Steene-Johannessen J, Brown WJ, et al. Does physical activity attenuate, or even eliminate, the detrimental association of sitting time with mortality? A harmonised meta-analysis of data from more than 1 million men and women. *Lancet.* 2016. doi:10.1016/S0140-6736(16)30370-1
8. Owen N, Sparling PB, Healy GN, Dunstan DW, Matthews CE. Sedentary behavior: emerging evidence for a new health risk. *Mayo Clin Proc.* 2010;85(12):1138-1141. doi:10.4065/mcp.2010.0444
9. Neuhaus M, Eakin EG, Straker L, et al. Reducing occupational sedentary time: A systematic review and meta-analysis of evidence on activity-permissive workstations. *Obes Rev.* 2014;15(10):822-838. doi:10.1111/obr.12201
10. Torbeyns T, Bailey S, Bos I, Meeusen R. Active workstations to fight sedentary behaviour. *Sports Med.* 2014;44(9):1261-1273. doi:10.1007/s40279-014-0202-x
11. Cao C, Liu Y, Zhu W, Ma J. Effect of Active Workstation on Energy Expenditure and Job Performance: A Systematic

- Review and Meta-analysis. *J Phys Act Heal*. 2016. doi:10.1123/jpah.2014-0565
12. MacEwen BT, MacDonald DJ, Burr JF. A systematic review of standing and treadmill desks in the workplace. *Prev Med (Baltim)*. 2015;70:50-58. doi:10.1016/j.ypmed.2014.11.011
 13. Heinrich A, Wasserkampf A, Schäfer A, Ellegast R, Kleinert J. Dynamic Office Workstations: Initial Insights into User Usability Evaluations and Motivation. *J UOEH*. 2018;40(4):277-286. doi:10.7888/juoeh.40.277
 14. Carr LJ, Leonhard C, Tucker S, Fethke N, Benzo R, Gerr F. Total Worker Health Intervention Increases Activity of Sedentary Workers. *Am J Prev Med*. 2016;50(1). doi:10.1016/j.amepre.2015.06.022
 15. Ghesmaty Sangachin M, Gustafson WW, Cavuoto LA. Effect of Active Workstation Use on Workload, Task Performance, and Postural and Physiological Responses. *IIE Trans Occup Ergon Hum Factors*. 2016;4(1):67-81. doi:10.1080/21577323.2016.1184196
 16. Karol S, Robertson MM. Implications of sit-stand and active workstations to counteract the adverse effects of sedentary work: A comprehensive review. *Work*. 2015;52(2):255-267. doi:10.3233/WOR-152168
 17. Commissaris DACM, Könemann R, Hiemstra-van Mastrigt S, et al. Effects of a standing and three dynamic workstations on computer task performance and cognitive function tests. *Appl Ergon*. 2014;45(6). doi:10.1016/j.apergo.2014.05.003
 18. Garland E, Watts A, Doucette J, Foley M, Senerat A, Sanchez S. Stand Up to Work: assessing the health impact of adjustable workstations. *Int J Work Heal Manag*. 2018;11(2):85-95. doi:10.1108/IJWHM-10-2017-0078
 19. Neuhaus M, Healy GN, Dunstan DW, Owen N, Eakin EG. Workplace sitting and height-adjustable workstations: A randomized controlled trial. *Am J Prev Med*. 2014;46(1). doi:10.1016/j.amepre.2013.09.009
 20. Healy GN, Eakin EG, LaMontagne AD, et al. Reducing sitting time in office workers: Short-term efficacy of a multicomponent intervention. *Prev Med (Baltim)*. 2013;57(1). doi:10.1016/j.ypmed.2013.04.004
 21. Grunseit AC, Chau JY, Van der Ploeg HP, Bauman A. Thinking on your feet: A qualitative evaluation of an installation of sit-stand desks in a medium-sized workplace. *J Sci Med Sport*. 2013;13(1):365. doi:10.1016/j.jsams.2012.11.479